COMSATS University Islamabad,

Sahiwal Campus.

Filia
Blockchain Video Streaming D-App

A Project Presented To

COMSATS University, Islamabad
Sahiwal Campus

In partial fulfillment

of the requirement for the degree of

Bachelor of Science in Software Engineering (2020-2024)

By

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The candidate confirms that the work submitted is their own and appropriate credit has been given where reference has been made to the work of other

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ZOHAIR KHAN **LAIBA ZULFIQAR**

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MUHAMMAD TALHA

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CERTIFICATE OF APPROVAL

It is to certify that the final year project of BS (SE) “Filia: Blockchain Video Streaming D-App” was developed by **Zohair Khan (CIIT/SP20-BSE-034/SWL)**, **Muhammad Talha (CIIT/SP20-BSE-049/SWL)** and **Laiba Zulfiqar (CIIT/SP20-BSE-039/SWL)** under the supervision of **Dr. Muhammad Farhan** that in their opinion; it is fully adequate, in scope and quality for the degree of Bachelor of Science in Software Engineering.

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Executive Summary

Filia is a groundbreaking decentralized video content platform, designed as an alternative to centralized streaming sites like YouTube and Netflix. It leverages blockchain technology and IPFS file storage to provide a secure and user-centric experience. Users can monetize their content, and the platform operates on Ethereum with smart contracts created using Polygon and Solidity. Filia promotes content freedom, creative expression, and a decentralized approach with no central authority, making it resistant to censorship. The platform's core objectives are data privacy and user empowerment. It addresses concerns about centralized platforms collecting and monetizing personal data. Filia offers a seamless and transparent video sharing experience, allowing users to control their content and access it securely through IPFS and Ethereum-based smart contracts. The integration of The Graph ensures efficient and scalable data querying. The project encompasses key features like user authentication, video upload and storage, content discovery, and an intuitive user interface. It follows an Agile development methodology to incorporate user feedback, ensuring a user-centric approach. Filia is a revolutionary decentralized video content platform that seeks to disrupt the landscape of traditional streaming sites like YouTube and Netflix. Unlike its centralized counterparts, Filia operates on blockchain technology, utilizing the Ethereum network, smart contracts, and Polygon for transparent transactions and content handling. This innovative approach empowers users by diluting central authority through the distribution of tokens, ensuring that content creators have equal ownership and control over their creations. The platform leverages an Interplanetary File System (IPFS) for file storage, guaranteeing the security of uploaded content. This not only safeguards user data but also provides resistance against censorship, creating a more liberated environment for creators and viewers alike. Filia's core mission is to provide a seamless, transparent, and decentralized video sharing experience while enabling content monetization for users.

Acknowledgment

All praise is to Almighty Allah who bestowed upon me a minute portion of His boundless knowledge by virtue of which I was able to accomplish this challenging task.

I am greatly indebted to my project supervisor **Dr. Muhammad Farhan** and co-supervisor **Mr. Ahmad Shaf**. Without their personal supervision, advice and valuable guidance, completion of this project would have been doubtful. I am deeply indebted to them for their encouragement and continual help during this work.

And we are also thankful to our parents and family who have been a constant source of encouragement for us and brought us the values of honesty & hard work.

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Abbreviations

IPFS	Interplanetary File System
API	Application Programming Interface
JS	JavaScript
SRS	Software Require Specification
SDD	Software Design Description
GUI	Graphical User Interface
Dapp	Decentralization App
IDEs	Integrated development environments
CSS	Cascading Style Sheet

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Chapter No. 1

Introduction

1 Introduction

Decentralized applications, or DApps, are essentially blockchain-based smart contract-powered versions of apps popularized by the Ethereum network. They act just like traditional apps — a user shouldn't even notice a difference — but provide much more in terms of a feature set. Filia is the intended blockchain app as the alternative for the likes of YouTube Netflix. The Dapp is a decentralized version of any major video streaming site. It will aim to provide content like any other video streaming site but through a blockchain architecture and IPFS file storage system. Users will be able to monetize their content. Central authority will be diluted by the distribution of tokens. It will run on an Ethereum wallet. Smart contract will be made through Polygon and Solidity through which all the transactions will be handled. User will be able to do all the functionalities as they can on other platforms but through a Peer-to-Peer networking and Interplanetary File System. It will be resistant to censorship. No annoying ads. More creative freedom is allowed. Altogether a more liberated and blockchain based version.

The Filia platform is a revolutionary video sharing application built on the Polygon network and The Graph, designed to address the growing concerns surrounding user privacy and content censorship. As an alternative to traditional video sharing platforms, Filia offers users a secure and decentralized environment to create, share, and watch videos without compromising their privacy. (ALLCHIN, JAMES EDWARD, 2007)

With the proliferation of online video content, users have become increasingly aware of the risks associated with centralized platforms that collect and monetize their personal data. Filia aims to tackle these issues by leveraging blockchain technology and the power of the Polygon network to ensure data privacy, content freedom, and user empowerment.

The core objective of Filia is to provide a seamless and transparent video sharing experience, allowing users to exercise full control over their content. By utilizing IPFS for file storage and the Ethereum-based smart contracts for transparent interactions, Filia enables users to securely upload, store, and access videos without the risk of data breaches or censorship.

The significance of Filia lies in its ability to reshape the online video sharing landscape. It introduces a decentralized approach that empowers users by putting them in control of their data and content. Additionally, the integration of The Graph allows for efficient and scalable data querying, ensuring a smooth user experience even as the platform grows.

The scope of the project encompasses the development and implementation of key features such as user authentication, video upload and storage, content discovery, and a seamless user interface. By adopting an Agile software development methodology, the project team will be able to iterate and adapt to user feedback, ensuring a user-centric approach throughout the development process.

In conclusion, Filia represents a paradigm shift in the world of video sharing platforms. By leveraging blockchain technology, decentralized storage, and user-centric design, Filia provides a safe, private, and censorship-resistant environment for users to share their content and engage with a global community. (Patel, Sethia, & Patil)

1.1 Brief Overview

Filia is a groundbreaking decentralized video content platform designed as an alternative to centralized streaming sites like YouTube and Netflix. It leverages blockchain technology, including Ethereum, Polygon, and Solidity, along with IPFS for secure file storage. Users can monetize their content, and Filia operates without central authority, promoting creative freedom and resisting censorship. The platform focuses on user privacy and content freedom, providing a seamless and transparent video sharing experience. The integration of The Graph ensures efficient data querying. The project follows Agile methodology, emphasizing a user-centric approach. In conclusion, Filia represents a paradigm shift in video sharing, offering a secure and censorship-resistant space for content creators and a global community of viewers.

1.2 Relevance to Course Modules

The Filia project aligns closely with a Bachelor's degree in Software Engineering by showcasing a comprehensive understanding of software development, blockchain technology, and smart contracts. It demonstrates the application of software engineering principles to create a decentralized video platform. Students pursuing this degree would gain valuable insights into blockchain, decentralized systems, user-centric design, and project management. This project offers a practical example of applying software engineering knowledge to develop innovative, secure, and user-centered software solutions. During our studies we also learnt knowledge like React, HTML, CSS, JS and we also learnt some extra technologies by short courses like Blockchain, truffle, Solidity (for writing smart contracts). (Naik, Vaity, D'mello, & Patil, 2017)

1.3 Project Background

Filia's genesis can be traced back to the inadequacies of traditional online video-sharing platforms like YouTube and Netflix. Prior to the advent of Web 3.0, these centralized platforms wielded complete control, monetizing user data and subjecting users to relentless ads. This centralization often led to content censorship and a lack of creative freedom.

As a response to these issues, Filia emerged as a pioneering Web 3.0 project. It harnesses the power of blockchain technology, Ethereum, Polygon, and Solidity for smart contracts to offer a decentralized alternative for video streaming. The Filia project prioritizes user privacy, empowers content creators to monetize their work, and eliminates central authority by distributing tokens. It operates on a peer-to-peer network, utilizing the secure Interplanetary File System (IPFS) for content storage.

The core objective of Filia is to provide a transformative video-sharing experience, where users have complete control over their content. By combining IPFS for content storage with Ethereum smart contracts for transparent transactions, Filia mitigates the risk of data breaches and censorship.

Filia stands as a testament to the evolution of online video sharing. It represents a paradigm shift from centralized control to user empowerment, with The Graph's integration ensuring efficient data querying as the platform grows. In essence, Filia is the answer to the limitations of the pre-Web 3.0 era, offering a secure, private, and censorship-resistant platform for content creators and a global community of viewers.

1.4 Analysis from Literature Review (in the context of your project)

Table 1.1 Literature Review of related applications

Application Name	**Weakness**	**Proposed Project Solution**
Filia Streaming Decentralized App	Vulnerability to Smart Contract Bugs.	Extensive code auditing and testing to identify and rectify potential vulnerabilities before deployment.

- The system is transparent because no centralized database system is used.
- Secure the contracts with cryptography and blockchain techniques.
- Identity of every user remain hidden from everyone.

1.4.1 Methodology and Software Lifecycle for This Project

This system will be developed into many development ways but most authentic and suitable life cycle is Incremental model and little bit Agile model. The incremental build model is a method of software development where the product is designed, implemented and tested incrementally until the product is finished. It involves both development and maintenance. The product is defined as finished when it satisfies all of its requirements.

1.4.2 Incremental Model

The Filia project follows an incremental development methodology, which involves breaking the project into smaller, manageable components. This approach allows for the gradual build-up of features and functionalities. Each increment represents a tangible improvement or addition to the system. In the context of Filia, this means that the development team focuses on specific features or modules in each iteration. For instance, they might begin with user authentication and then incrementally add video upload and storage, content discovery, and other key features. This incremental approach ensures that the project can adapt to changing requirements and allows for early testing and user feedback on individual components.

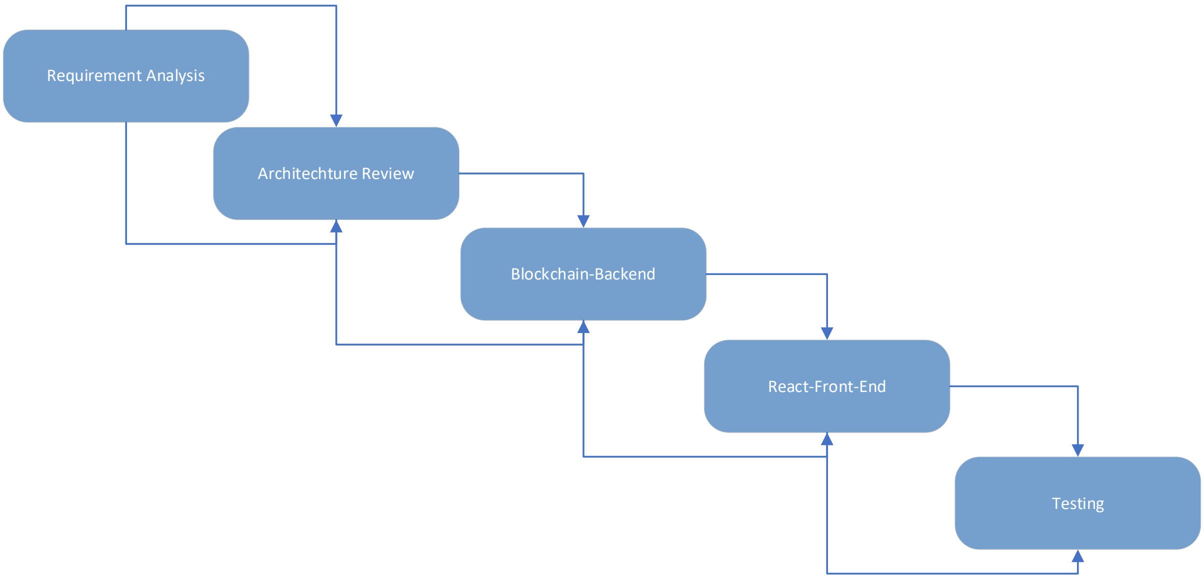

Figure 1.1 Filia Incremental Model Diagram

1.4.3 Agile Model

The Filia project also adopts an Agile methodology, emphasizing collaboration, flexibility, and continuous improvement. Agile methodologies promote iterative development, with a strong focus on responding to user feedback and adjusting the project's direction accordingly. In the case of Filia, Agile principles guide the development team in staying responsive to user needs and evolving project requirements. They regularly solicit and incorporate user feedback to ensure that the platform remains user-centric. Agile practices such as daily stand-up meetings, sprint planning, and frequent testing are integral to Filia's development process, ensuring that it remains adaptable and aligned with user expectations.

Chapter No. 2

Problem Definition

2 Problem Definition

This chapter discusses the precise problem to be solved. It should extend to include the outcome.

2.1 Problem Statement

Traditional video-sharing platforms like YouTube suffer from several limitations, including centralized control over user data, lack of privacy, and vulnerability to censorship. Users have little control over their own content and are subject to arbitrary takedowns or demonetization. Additionally, content creators face challenges in monetizing their videos and lack transparency in revenue sharing. There is a need for a decentralized video-sharing platform that empowers users, ensures data privacy, and provides transparent and fair monetization opportunities. Filia represent a way of video streaming through Blockchain smart contract. When one thinks of traditional streaming, uploading, browsing, streaming, and searching similar entities come to mind. Each of these is powered by a central authority such as Netflix, YouTube, Hulu. We are intending to remove that central authority by diluting it. It is a major problem in platform that the organization can remove any individuals channel and content without any reason and speak. On our Blockchain data cannot be deleted only re-edited because of P2P. Also, they censorship content which disrupts creativity. Decentralization removes authority so no ban or censorship. We want to learn about Blockchain, Web3, DApps and other services and for a decentralized solution of censorships and ads. Vimeo and Dailymotion exist but they are not blockchain based. We opt to learn about Dapps and Blockchain by implementing this concept. We hope to learn Next.js for front end, Solidity for making contracts and transactions, IPFS Querying data: The Graph, Layer 2 blockchain: Polygon. (Ashlesh Gawande, 2012)

2.2 Deliverables and Development Requirements

In the context of the Filia project, deliverables refer to the tangible outcomes, features, and components that will be developed and provided as part of the project.

2.2.1 Deliverables

User Authentication System: Developing a secure and user-friendly system for user authentication, ensuring the privacy and security of user data by using Meta Mass wallet.

- **Video Upload and Storage:** Creating a seamless mechanism for content creators to upload videos to the platform and securely store them using the Interplanetary File System (IPFS).
- **Content Discovery:** Implementing algorithms and features that allow users to discover

and explore a wide range of videos and content based on their interests and preferences.

- **Smart Contract Development:** Designing and implementing smart contracts using Solidity and Polygon for handling transactions, token distribution, and other blockchain-related functionalities.
- **User Interface (UI) Design:** Designing an intuitive and user-friendly interface that enhances the overall user experience and encourages engagement with the platform.

2.2.2 Development Requirements

Development requirements are the conditions, tools, and technologies necessary for the successful execution of the project. In the case of Filia, these requirements encompass:

- **Blockchain Expertise:** Developers must possess a strong understanding of blockchain technology, Ethereum, Polygon, and Solidity to create and manage smart contracts and ensure the platform's blockchain integration.
- **Interplanetary File System (IPFS):** Proficiency in utilizing IPFS for secure content storage and retrieval is crucial to safeguard user data and ensure content availability.
- **Agile Development Practices:** Following Agile methodologies such as Scrum or Kanban to manage project tasks, foster collaboration, and adapt to changing requirements.
- **Security Auditing:** Conducting thorough security audits to identify and mitigate vulnerabilities, ensuring the platform's resilience to potential threats.
- **User-Centric Approach:** Prioritizing user feedback and continuous improvement to create a platform that resonates with user expectations and needs.

2.3 Current System

The current system for the Filia project represents a significant milestone in the development process. At this stage, the project has successfully completed the primary features and is deployed on the Vercel platform, providing a live and accessible environment for users. One of the noteworthy accomplishments is the integration of MetaMask wallet for user authorization, which adds an essential layer of security and convenience to the platform.

With the primary features in place, users can now securely create accounts, upload their video content, explore a wide array of videos, and interact with the platform in a decentralized manner. The platform leverages the Ethereum network and its associated wallet, MetaMask, to ensure secure and efficient authorization. MetaMask, a popular cryptocurrency wallet, not only offers a reliable method for user authentication but also facilitates seamless transactions and interactions within the blockchain-based environment.

The deployment on Vercel ensures that the Filia platform is readily accessible to users, providing a smooth and responsive user experience. This achievement marks a significant step towards the platform's mission to empower content creators and viewers with a decentralized, secure, and user-centric video-sharing ecosystem. It also positions Filia as a promising contender in the evolving landscape of online video platforms.

As the project progresses, the development team will continue to iterate and enhance the platform based on user feedback, further solidifying Filia's position as a paradigm shift in video sharing, where user privacy, content freedom, and decentralization are at the forefront of the user experience.

Chapter No. 3

Requirement Analysis

3 Requirement Analysis

3.1 Use Case Diagrams

In this Use case diagrams, we show the behavior of our system and frequently use it to analyze various systems. They enable you to visualize the different types of roles in a system and how those roles interact with the system.

Figure 3.1 Filia D-App Multiple Use Cases Diagram

3.2 Use Case Descriptions

Table 3.1 Select Wallet Use Case

Use Case	**Select Wallet**
ID	UC1
Description	User will upload video without showing his/her identity
Primary Actors	User
Secondary Actors	None
Preconditions	None
Main Flow	The use case starts when the user clicks the "Wallet" tab on the navigation bar. They will see different Option of wallets. The system displays the different wallet option that allows users to select Wallet and enter to website without showing Identity. The system displays the user's profile page.
Post Conditions	User enter in the system
Alternative Flows	Enter as Guest

Table 3.2 Select Leader Use Case

Use Case	**Select Leader**
ID	UC2
Description	User will select leader by their Voting
Primary Actors	Companies, academics, and students
Secondary Actors	None
Preconditions	Connect with MetaMask Wallet
Main Flow	User enter in system by connecting with Wallet then User is able to Vote for the leader Select to performed some specific tasks.
Post Conditions	Follow the Leader Guidelines for certain tasks.
Alternative Flows	None

Table 3.3 Join as Guest Use Case

Use Case	Join as Guest
ID	UC3
Description	User tries to enter into the system without wallet connection
Primary Actors	Companies, academics and students
Secondary Actors	None
Preconditions	None
Main Flow	The use case starts when the user clicks the "Join as Guest" tab on the navigation bar. The system displays the welcome page (Screen) but user do not
Post Conditions	The user has access to the system
Alternative Flows	MetaMask Wallet Connection

Table 3.4 Comment on Video Use Case

Alternative Flow	Comment on Video
ID	UC4
Description	User is able to comment on Videos
Primary Actors	Companies, academics and students
Secondary Actors	None
Preconditions	Enter as Guest or MetaMask Wallet
Main Flow	User is entering as guest or MetaMask wallet and able to comment on Video User play video then comment on it.
Post Conditions	None

Alternative Flows	None

Table 3.5 Feedback Use Case

Use Case	**Feedback**
ID	UC5
Description	User is able to Feedback on website and give suggestions
Primary Actors	User
Secondary Actors	None
Preconditions	Enter as Guest or MetaMask Wallet
Main Flow	User is entering as guest or MetaMask wallet and able to give Feedback on Websites and Suggestions
Post Conditions	None
Alternative Flows	None

Table 3.6 Search for Video Use Case

Use Case	**Search for Video**
ID	UC6
Description	User wants to search for a video that they are interested
Primary Actors	User
Secondary Actors	None
Preconditions	Enter as Guest or MetaMask Wallet
Main Flow	The use case starts when the user clicks the search bar The system displays the page with all kinds of relevant videos The user clicks on the “Search” button and type in the relevant skill(s) as keywords The system displays a list of videos that matched the skill(s) input by user The user clicks on the video that they are interested

Table 3.7 Streaming Video Use Case

Use Case	**Stream Video**
ID	UC7
Description	Users want to play a video
Primary Actors	User
Secondary Actors	None
Preconditions	Enter as Guest or MetaMask Wallet
Main Flow	The use case starts when the user clicks a video The video starts to play and buffer
Post Conditions	A project is created and the details are stored in the database
Alternative Flows	None

Table 3.8 Upload Video Use Case

Use Case	**Upload Video**
ID	UC8
Description	Content Creator user wants to upload a video on site
Primary Actors	Content Creator user
Secondary Actors	None
Preconditions	Connect to MetaMask Wallet
Main Flow	The use case starts when the user clicks the "My videos" tab on the navigation bar. The system displays the project page with all kinds of available videos. The user clicks on the "Upload Video" button The system displays a page that allow user to upload video

	The user fills in the details and and upload video and click on the "Upload" button
Post Conditions	A project is created and the details are stored in the database(P2P)
Alternative Flows	None

3.3 Functional Requirements

3.3.1 User Login.

- The system shall allow users to create an account on metamask

3.3.2 Video Upload:

- The system shall allow authenticated users to upload videos to the platform.
- The system shall support video file formats such as MP4, AVI, and MOV.
- The system shall validate the size and format of the uploaded videos.
- The system shall store the uploaded videos in a secure and reliable storage system.

3.3.3 Video Management:

- The system shall provide functionality for users to manage their uploaded videos.
- Users shall be able to edit video titles, descriptions, and other metadata.
- Users shall be able to delete their own videos from the platform.
- Users shall be able to set privacy settings for their videos (public, private, unlisted).

3.3.4 Video Playback:

- The system shall allow users to view and play videos on the platform.
- The system shall support video streaming to ensure smooth playback.
- Users shall be able to control video playback, such as play, pause, seek, and volume control.
- The system shall provide options for video quality and resolution selection based on the user's internet connection.

3.3.5 Video Search and Discovery:

- The system shall provide a search functionality for users to find videos based on keywords, titles, or tags.

- The system shall display relevant search results to users in a user-friendly manner.
- The system shall offer video recommendations and suggested videos based on user preferences and viewing history.

These functional requirements outline the core functionalities and features of the Filia application. You can further expand on these requirements or add more specific requirements based on the needs and goals of your project.

3.4 Non-functional Requirements

Nonfunctional requirements are those that establish parameters that may be used to evaluate the performance of a system. Nonfunctional requirements are the limitations to which the system will work. Language execution environment, operating environment performance parameters, usability requirements, and so on. These are all the needs that do not fall under functional requirements but impact on the system.

3.4.1 Usability

The system will assist the user in locating and streaming videos in the shortest amount of time. The consumer can choose to upload videos on their channel. The system will be user friendly and easy to understand.

3.4.2 Performance

The system's performance is excellent. It completes all tasks correctly and offers all results fast and precisely. All the work will be decentralized every transaction have different block in blockchain.

3.4.3 Accessibility

Anyone can access this app, but the main functionalities can only be accessed by logged in users. To just see the video, you do not need to log in but if you want to upload the video edit it then you must log in to the system.

3.4.4 Responsiveness

The system will be very user-friendly in such a way that it follows the users’ instructions and the response on time. The system responds to every user error. When a user uses a service, this system immediately sends the client's information to the service provider.

3.4.5 Availability

This web application will be available to all users at all times, everyone can use any functionality of this decentralized web site anytime of the day and chances of one point failure

Chapter No. 4

Design and Architecture

4 Design and Architecture

4.1 System Architecture

The structure of the system explains the working of the system and describes the components of the project, their relationships, and how they interact with each other, with this architecture.

4.2 Process Flow/Representation

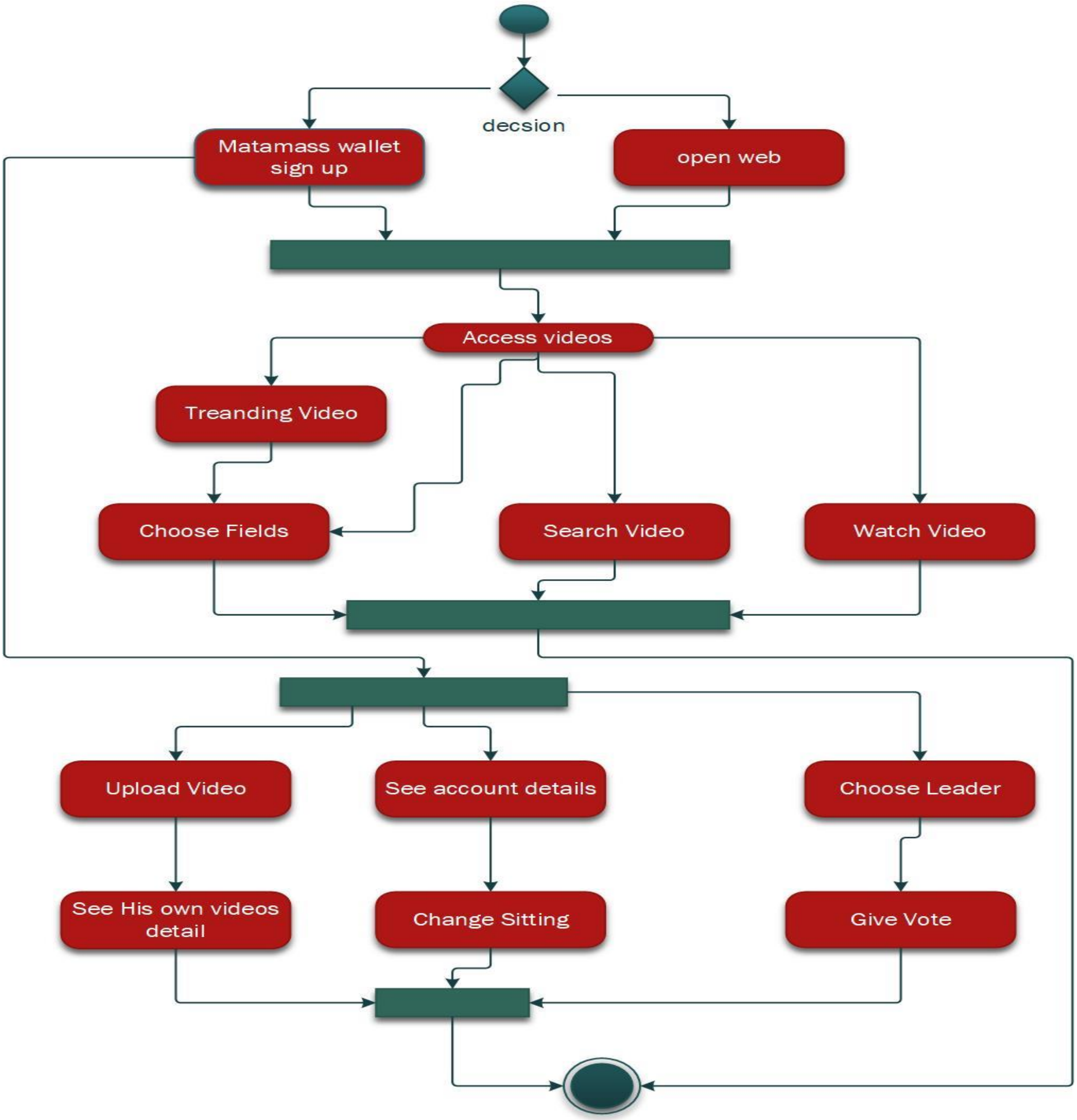

Figure 4.1 Filia D-App Data Flow / Control Flow diagram

4.3 Sequence Diagrams

The sequence diagram is a visual representation of the interactions between the different objects and components of the system. It shows the flow of messages between the objects and how they collaborate to achieve specific tasks in the system. The sequence diagram captures the dynamic behavior of the system and is useful in identifying potential issues and optimizations in the system's design

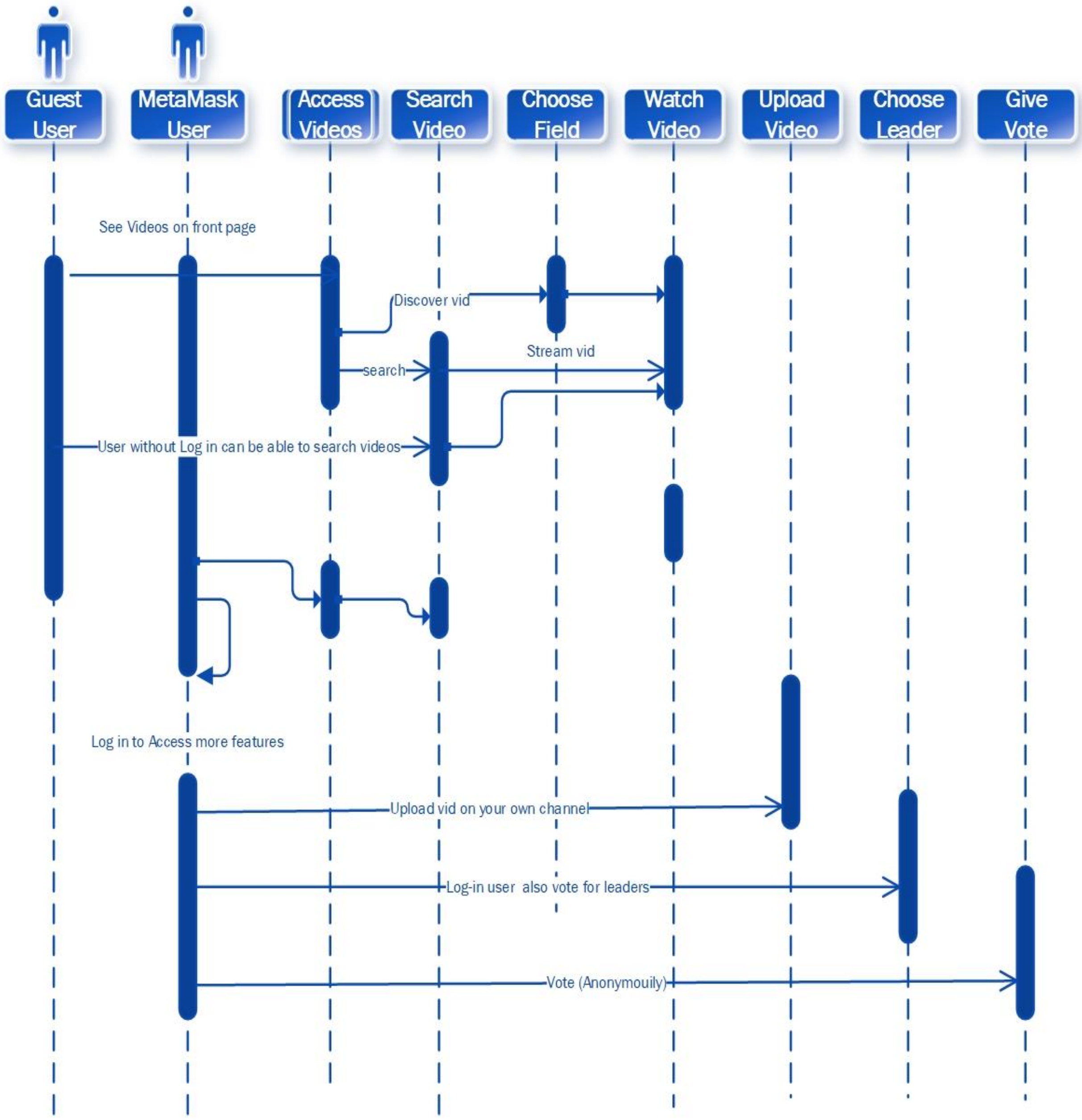

Figure 4.2 Filia Video Upload and Streaming Sequence Diagram

4.4 Class Diagram

The class diagram is a graphical representation of the system's classes, their attributes, and the relationships between them. It depicts the static structure of the system and shows how the different classes in the system are related to each other.

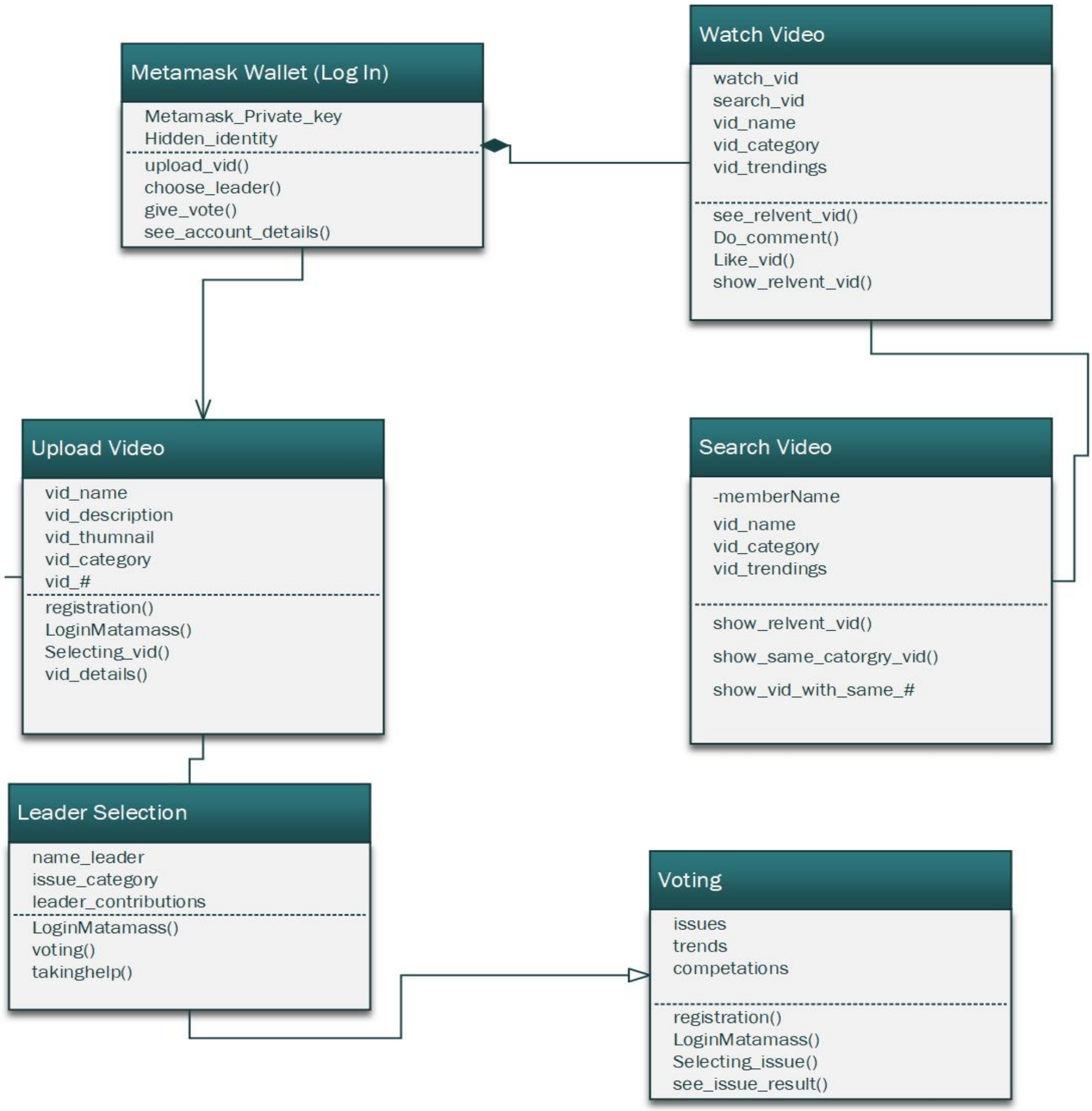

Figure 4.3 Filia Voting & Authentication Class Diagram

Chapter No. 5
Implementation

5 Implementation

The implementation of the Filia project involves a complex set of technologies and processes to create a decentralized video content platform. Key components include the use of blockchain technology, the Ethereum network, smart contracts, IPFS for content storage, and integration with the MetaMask wallet. Below is an overview of the project's implementation and some high-level algorithms involved:

- **Blockchain Integration**

  Filia relies on the Ethereum blockchain for decentralized operations. Smart contracts written in Solidity govern various functionalities, such as token distribution, content monetization, and user management.

- **IPFS Integration**

  The Interplanetary File System (IPFS) is used for secure and decentralized content storage. Videos are uploaded to IPFS, and their content hashes are recorded on the blockchain.

- **MetaMask Wallet**

  MetaMask is integrated to handle user authorization and cryptocurrency transactions. Users connect their wallets to Filia for secure access and content monetization.

- **User Management**

  Users can register, authenticate, and edit their profiles. Smart contracts maintain user data and access control.

- **Content Discovery**

  Algorithms are implemented to recommend and display relevant content based on user preferences and interaction history.

- **Content Monetization**

  Filia employs smart contracts to handle token distribution and rewards for content creators when users access their videos.

5.1 Algorithm

Content Recommendation Algorithm

Filia's content discovery algorithm analyzes user behavior and preferences to recommend videos. It employs collaborative filtering, content-based filtering, or a hybrid approach to suggest content.

Token Distribution Algorithm

Smart contracts execute token distribution for content creators based on views, likes, and other engagement metrics. The algorithm calculates token rewards for each video.

Video Playback Algorithm

To ensure efficient video playback, an algorithm fetches video data from IPFS and optimizes streaming to reduce buffering and enhance the user experience.

Security Algorithms

Various security algorithms, including encryption and validation checks, protect user data, smart contracts, and content storage from potential threats and vulnerabilities.

Performance Optimization

Algorithms are in place to monitor and optimize the platform's performance, including load balancing, caching, and scaling based on user traffic.

Monetization and Payment Algorithm

Smart contracts handle the collection of payments from users and the distribution of earnings to content creators based on specific monetization agreements.

The Filia project's implementation encompasses the integration of multiple technologies and the deployment of various algorithms to deliver a decentralized video-sharing platform that ensures user privacy, content freedom, and security while promoting user engagement and empowerment.

5.2 User Interface

Below Figure 5.1, Figure 5.2, Figure 5.3 represent the user interfaces and screens of Filia Blockchain based decentralized application with the landing page and home screen and upload page.

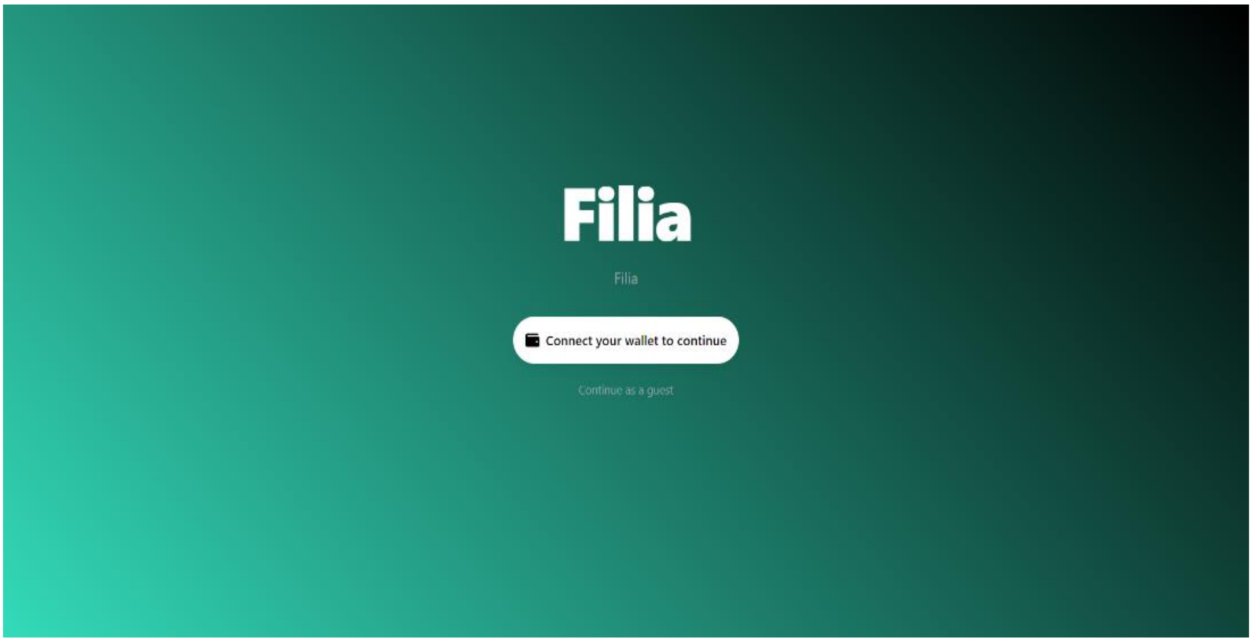

Figure 5.1 Filia Landing Page

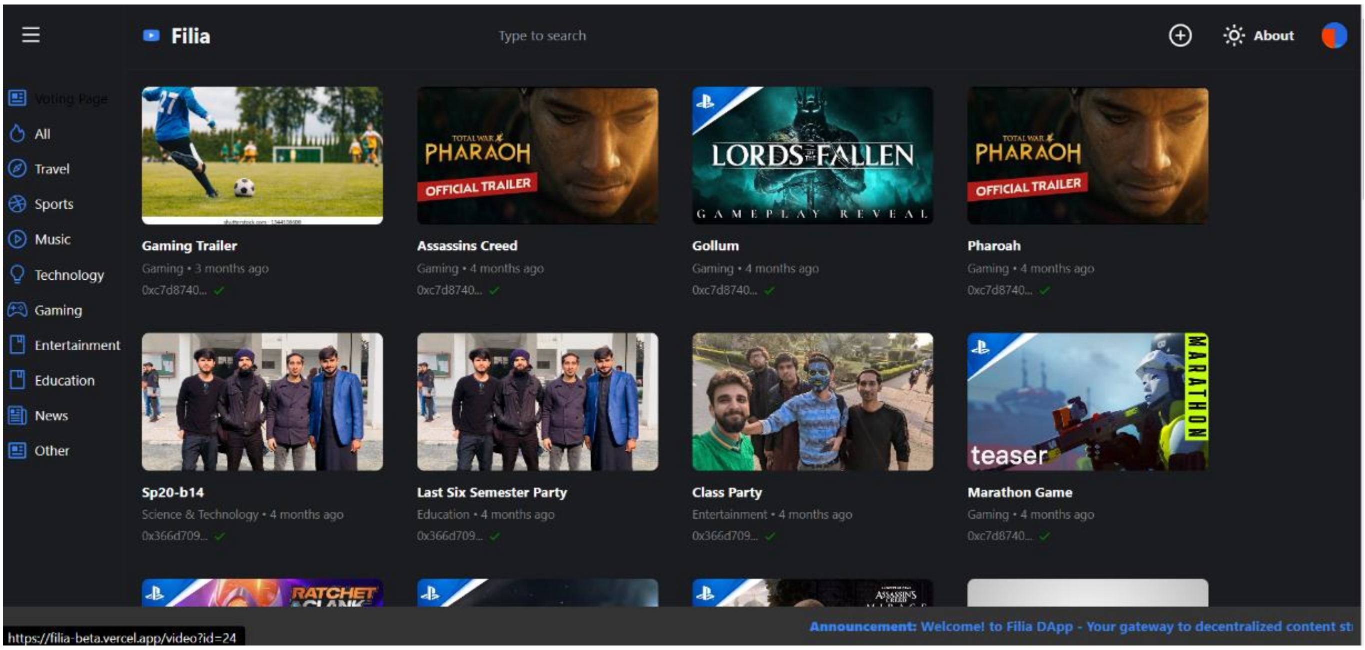

Figure 5.2 Filia Home Screen

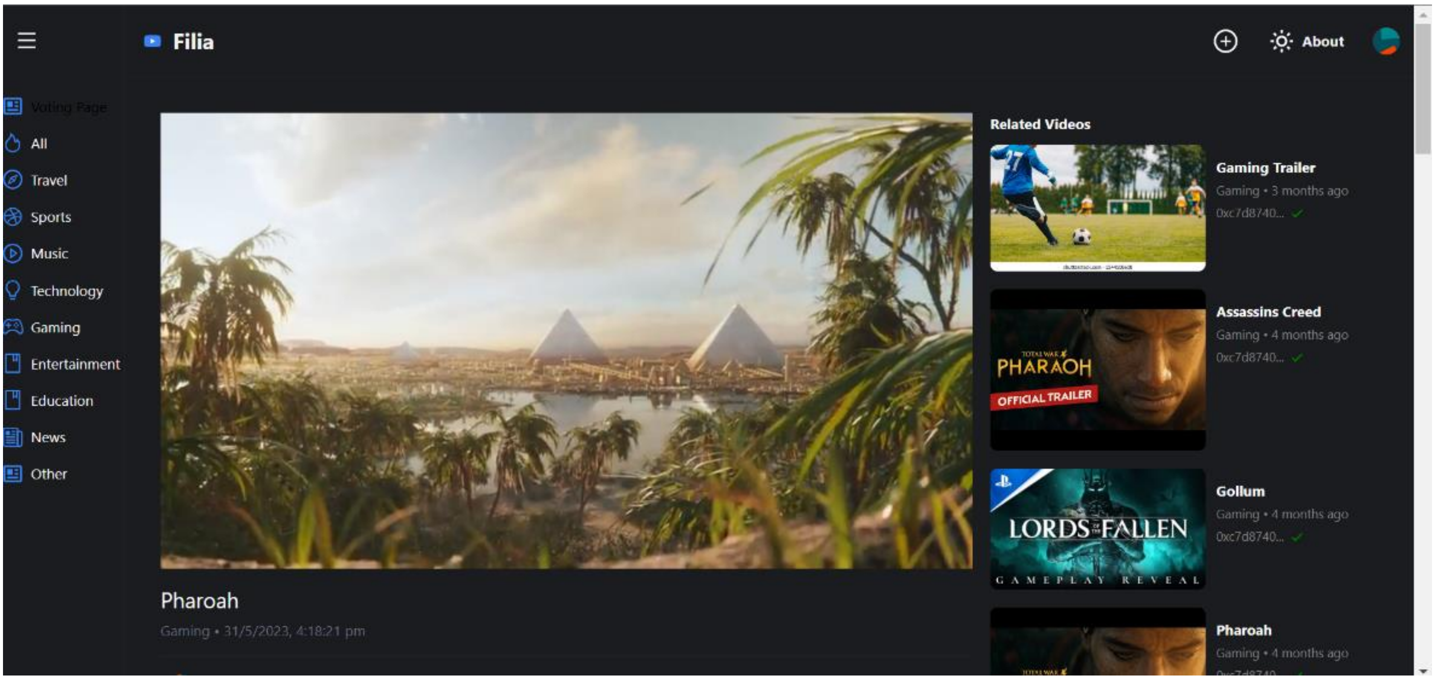

Figure 5.3 Filia Video Streaming Screen

Chapter 6

Testing and Evaluation

6 Testing

Testing and evaluation are critical phases in the development of the Filia project to ensure that the platform operates as expected, is secure, and provides an excellent user experience. Here, we outline the key aspects of testing and evaluation:

6.1 Unit Testing

Unit tests are conducted to assess the individual components of the Filia platform, such as smart contracts, MetaMask wallet integration, and IPFS functionality. These tests focus on verifying the correctness and reliability of each component.

Table 6.1 Unit Test Cases

Test Case ID	**Test Case Name**	**Attributes & Values**	**Expected Result**	**Result**
UTC001	Smart Contract	- Deploy and interact with smart contracts	Successful contract deployment and interaction	Pass
UTC002	IPFS Integration	- Upload and retrieve content to/from IPFS	Successful content storage and retrieval	Pass
UTC003	MetaMask Wallet	- Connect MetaMask wallet to Filia	Successful wallet connection	Pass
UTC004	Token Distribution	- Distribute tokens for content monetization	Correct token distribution	Pass

6.2 Manual Testing

Manual testing is crucial for evaluating user-centric functionalities. Testers interact with the platform through a user interface to ensure that user registration, video upload, content discovery, and monetization features work as intended. This phase also includes testing the MetaMask wallet integration to ensure seamless authorization.

Table 6.2 Manual Test Cases

Test Case ID	Test Case Name	Attributes & Values	Expected Result	Result
MT001	User Registration	- Register with MetaMask	Registration is successful	Pass
MT002	Video Upload	- Upload a video with MetaMask authorization	Video is uploaded and associated with user	Pass
MT003	Content Discovery	- Search for videos and view with MetaMask	Relevant videos displayed, MetaMask authorization required	Pass
MT004	Comment Interaction	- Post a comment with MetaMask authorization	Comment is successfully posted	Pass
MT005	User Profile Management	- Edit profile info with MetaMask authorization	Profile info is updated	Pass

MT006	Monetization	- Associate wallet with MetaMask for content monetization	Wallet linked; content is monetized	Pass
MT007	Video Deletion	- Delete a video with MetaMask authorization	Video is removed	Pass

6.3 Functional Testing

Functional testing examines the system's functionality as a whole. It includes exploring all primary features, such as user authentication, content discovery, and interaction. The system's ability to maintain privacy, security, and responsiveness is assessed.

Table 6.3 Functional Test Cases

Test Case ID	Test Case Name	Attributes & Values	Expected Result	Result
FT001	System Functionality	- Explore all primary features	All features work as intended	Pass
FT002	Security and Privacy	- Attempt unauthorized access, privacy checks	Unauthorized access is denied, privacy measures are effective	Pass
FT003	Compatibility	- Test on different browsers, devices	Filia works across various browsers and devices	Pass

FT004	Performance Testing	- Simulate heavy traffic, uploads	System maintains performance under heavy load	Pass

6.4 Security and Privacy Testing

Filia undergoes rigorous security and privacy testing. Security audits are performed to identify and rectify vulnerabilities in smart contracts and other components. Additionally, privacy checks ensure that user data is adequately protected.

6.5 Compatibility Testing

Compatibility testing is conducted to confirm that Filia is accessible across various web browsers and devices, including desktop and mobile platforms. This ensures a consistent user experience.

6.6 Performance Testing

Performance testing evaluates the platform's response to heavy user traffic and content uploads. The system's ability to maintain performance under a heavy load is assessed to provide a seamless experience to users.

6.6.1 Evaluation Metrics

During testing, the Filia project is evaluated using specific metrics and key performance indicators (KPIs) to determine its success and effectiveness:

- **User Engagement:** Metrics such as user registrations, content uploads, and user interactions (likes, comments, shares) are tracked to assess the level of user engagement with the platform.
- **Security and Privacy:** Security audits are conducted to identify and address vulnerabilities, and the platform's ability to protect user data is evaluated. The number of security issues resolved is a critical metric.
- **Performance Metrics:** Performance is evaluated based on response times, page load times, and system scalability. The platform's ability to handle concurrent user activities is assessed.
- **Content Monetization**: The success of content monetization is assessed by monitoring the number of content creators who receive earnings through smart contracts.
- **User Feedback**: User feedback and satisfaction metrics, collected during testing and from

the initial user base, play a significant role in evaluating the platform's usability and effectiveness.

- **Compatibility:** The platform's compatibility with various web browsers and devices is evaluated based on the percentage of successful interactions on different platforms.

Chapter No. 7

Conclusion and Future Work

7 Conclusion and Future Work

7.1 Conclusion

The Filia project represents a transformative leap in the realm of video-sharing platforms. By leveraging blockchain technology, decentralized storage, and user-centric design, Filia offers a safe, private, and censorship-resistant environment for content creators and viewers. The platform successfully integrates the Ethereum network, smart contracts, IPFS, and the MetaMask wallet for secure authorization.

Throughout the project's development, rigorous testing and evaluation were conducted to ensure the platform's functionality, security, and performance. User engagement metrics, content monetization, and positive user feedback are indicators of Filia's early success.

Filia's implementation demonstrates its ability to empower users, providing a secure and innovative platform for sharing and monetizing video content. The project's unit testing, manual testing, functional testing, security assessments, compatibility checks, and performance evaluations have been instrumental in shaping Filia's evolution.

As Filia moves forward, the project aims to further enhance user experiences, expand features, and scale its infrastructure to accommodate a growing user base. The project's commitment to user feedback and continuous improvement ensures that it remains at the forefront of the Web 3.0 era, redefining the landscape of online video sharing.

7.2 Future Work

In the future, the Filia project has several avenues for further development and improvement:

Enhanced Features: Expanding the platform's feature set to include live streaming, additional content discovery mechanisms, and improved user interaction tools.

Decentralization: Strengthening the decentralization aspect of Filia by exploring more advanced blockchain technologies and governance models.

Community Building: Fostering an active and engaged user community by introducing user-generated content challenges, incentives, and interactive events.

Mobile Applications: Developing dedicated mobile applications to provide a seamless experience on various mobile devices.

Accessibility: Ensuring accessibility for a wider audience by providing support for multiple languages and enhancing the user interface for people with disabilities.

Scalability: Continuously optimizing the platform's infrastructure to handle increased user traffic and content uploads.

Educational Initiatives: Providing resources and educational content on blockchain technology and content monetization for content creators and users.

Content Quality Control: Implementing advanced content moderation mechanisms and algorithms to ensure the quality and safety of content on the platform.

The Filia project remains committed to its mission of empowering users to share and monetize their content in a decentralized and secure environment. Future work focuses on continuous improvement, innovation, and furthering the platform's impact on the world of online video sharing.

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